

Design and Implementation of a Quadrature Down Converter Using Op-Amp Oscillator, NMOS Mixer, and RC Low-Pass Filter

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Abstract—This report presents the design, simulation, and laboratory implementation of a quadrature down converter (QDC). The system consists of a quadrature oscillator, two NMOS mixer stages, low-pass filters, and final output conditioning stages. The quadrature oscillator generates two sinusoidal local oscillator signals having nearly equal amplitude and approximately 90° phase separation. These signals are mixed with the input signal using MOSFET switch-based mixers to generate intermediate-frequency in-phase and quadrature components. A passive RC low-pass filter with a cut-off frequency close to 3 kHz is used to reject high-frequency mixer products and retain the required difference-frequency component. The complete prototype is verified using LTspice simulations and laboratory measurements. Results show successful generation of I-Q outputs, observable frequency translation, and practical agreement between simulation and hardware within expected tolerances caused by device non-idealities, breadboard parasitics, and component tolerances.

Index Terms—Quadrature down converter, I-Q mixing, NMOS mixer, quadrature oscillator, low-pass filter, intermediate frequency, LTspice, UA741.

I. INTRODUCTION

Modern wireless receivers usually receive signals at high radio frequencies, while baseband processing and filtering are easier at much lower frequencies. A down converter translates the received input signal from a high input frequency f_{IN} to an intermediate frequency f_{IF} . In a quadrature down converter, the input is mixed with two local oscillator signals separated by 90° . The two output paths are called the in-phase path, or I path, and the quadrature path, or Q path.

The input signal can be written as

$$v_{in}(t) = A_1 \cos(\omega_{IN}t). \quad (1)$$

The two oscillator outputs are

$$v_{OSCI}(t) = A_2 \cos(\omega_{OSC}t), \quad (2)$$

$$v_{OSQ}(t) = A_2 \sin(\omega_{OSC}t). \quad (3)$$

After multiplication, the I and Q mixer outputs become

$$v_{IFI} = \frac{A_1 A_2}{2} [\cos((\omega_{IN} - \omega_{OSC})t) + \cos((\omega_{IN} + \omega_{OSC})t)], \quad (4)$$

$$v_{IFQ} = \frac{A_1 A_2}{2} [\sin((\omega_{IN} + \omega_{OSC})t) - \sin((\omega_{IN} - \omega_{OSC})t)]. \quad (5)$$

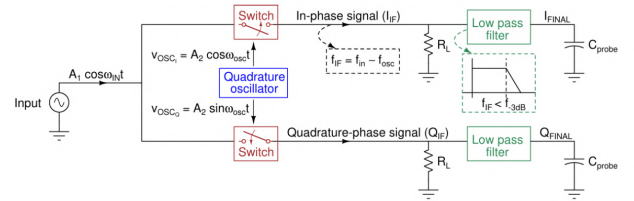


Fig. 1: Block diagram of the quadrature down converter.

Thus, mixing produces both sum and difference frequency components. The low-pass filter removes the high-frequency sum term and preserves the desired intermediate frequency

$$f_{IF} = |f_{IN} - f_{OSC}|. \quad (6)$$

II. NEED FOR QUADRATURE I-Q OPERATION

Quadrature operation is useful because a single real mixer loses sign information about whether the input frequency is above or below the local oscillator. For example, if $f_{IN} = f_{OSC} + \Delta f$ and another signal appears at $f_{IN} = f_{OSC} - \Delta f$, both produce the same absolute intermediate frequency Δf after ordinary real mixing. These two frequencies are called image frequencies. Without I-Q processing, the receiver cannot distinguish between them only from the magnitude of the IF signal.

I-Q down conversion solves this by preserving both amplitude and phase information. The two local oscillator signals form an orthogonal basis: one cosine component and one sine component. The complex representation of the received signal can be written as

$$x(t) = I(t) + jQ(t). \quad (7)$$

After quadrature mixing and low-pass filtering, the complex baseband signal contains the sign of the frequency offset. A positive frequency offset and a negative frequency offset rotate in opposite directions in the complex I-Q plane. Therefore, the receiver can identify whether the original signal was above or below the local oscillator.

Quadrature reception is also essential in modern digital modulation schemes such as QPSK, QAM, Wi-Fi, Bluetooth, and WLAN systems. These modulation schemes encode information in both amplitude and phase. The I path captures

the cosine projection of the signal, while the Q path captures the sine projection. Together, they allow full recovery of the complex symbol.

Another important advantage is image rejection. If the received RF signal is represented using complex exponentials, then multiplication by a complex local oscillator gives

$$x(t)e^{-j\omega_{osc}t}. \quad (8)$$

This shifts the desired channel to baseband while keeping the sign of the frequency. As a result, later digital signal processing can separate desired and image components more effectively than a single real mixer.

III. QUADRATURE OSCILLATOR DESIGN

A. Topology

The quadrature oscillator was implemented using a two-integrator loop topology with UA741 op-amps. The first op-amp stage behaves as an inverting integrator and the second op-amp stage behaves as a non-inverting integrator. The two outputs are fed back in a closed loop, resulting in two sinusoidal signals that are approximately 90° apart in phase.

The first stage transfer function is

$$H_1(s) = -\frac{1}{sR_1C_1}. \quad (9)$$

The second stage transfer function is

$$H_2(s) = \frac{1}{sR_3C_3}. \quad (10)$$

The loop gain is therefore

$$L(s) = H_1(s)H_2(s) = -\frac{1}{s^2R_1C_1R_3C_3}. \quad (11)$$

At $s = j\omega$, the loop phase becomes zero and the magnitude condition gives

$$\omega_0 = \frac{1}{\sqrt{R_1C_1R_3C_3}}, \quad (12)$$

$$f_0 = \frac{1}{2\pi\sqrt{R_1C_1R_3C_3}}. \quad (13)$$

Back-to-back 1N4148 diodes with small series resistors were used for amplitude stabilization. The diodes limit excessive growth of oscillation amplitude and help obtain an output amplitude close to 1 V_{PP} .

B. Design Calculations and Component Values

The target oscillator frequency was approximately 170 kHz. The ideal time constant is

$$\tau = \frac{1}{2\pi f_0} = \frac{1}{2\pi(170 \times 10^3)} \approx 0.936 \mu\text{s}. \quad (14)$$

Using capacitors in the nF and pF range, the resistor values were tuned in LTspice to account for UA741 bandwidth limitations, phase shift, and practical component availability.

TABLE I: Quadrature Oscillator Component Values

Component	Value	Function
R_1	1.7 k $\blacksquare$	U1 integrator input resistor
C_1	100 pF	U1 feedback capacitor
R_2	740 $\blacksquare$	U2 non-inverting input resistor
C_2	1 nF	U2 RC input capacitor
R_3	750 $\blacksquare$	U2 bias resistor
C_3	1 nF	U2 feedback capacitor
D1–D4	1N4148	Amplitude limiting diodes
Series resistors	100 $\blacksquare$	Diode current limiting
Supply	$\pm 12\text{ V}$	Op-amp supply

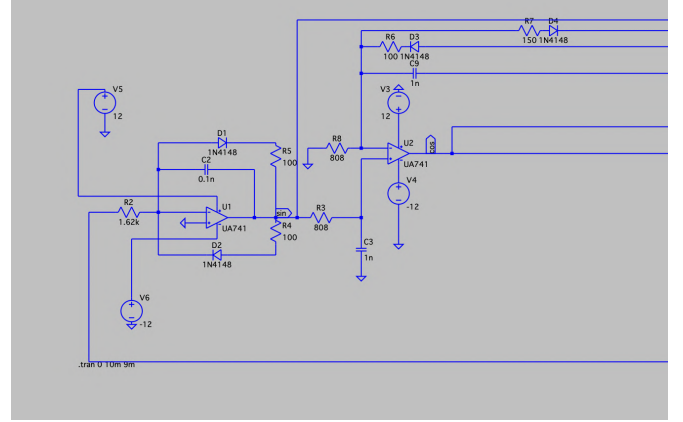


Fig. 2: LTspice schematic of the quadrature oscillator.

TABLE II: Quadrature Oscillator Simulation Summary

Parameter	Specification	Simulated
Oscillation frequency	170 kHz	169.3 kHz
v_{OSCI} amplitude	1 V _{PP}	1.05 V _{PP}
v_{OSCO} amplitude	1 V _{PP}	0.99 V _{PP}
Phase difference	90°	91°
Supply voltage	$\pm 12\text{ V}$	$\pm 12\text{ V}$

C. LTspice Simulation Results

The oscillator was simulated using the provided UA741 model file. The transient directive used was

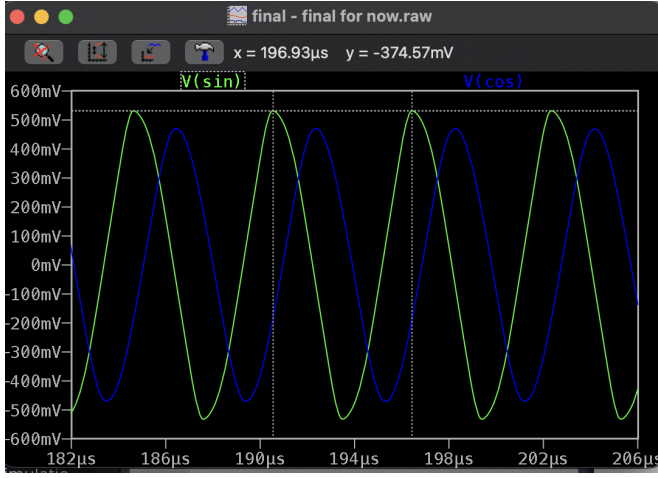
```
.tran 0 20m 15m 10n uic
.ic V(VOSCI)=0.1
.include UA741.301
```

The initial condition was used only to start oscillation in simulation.

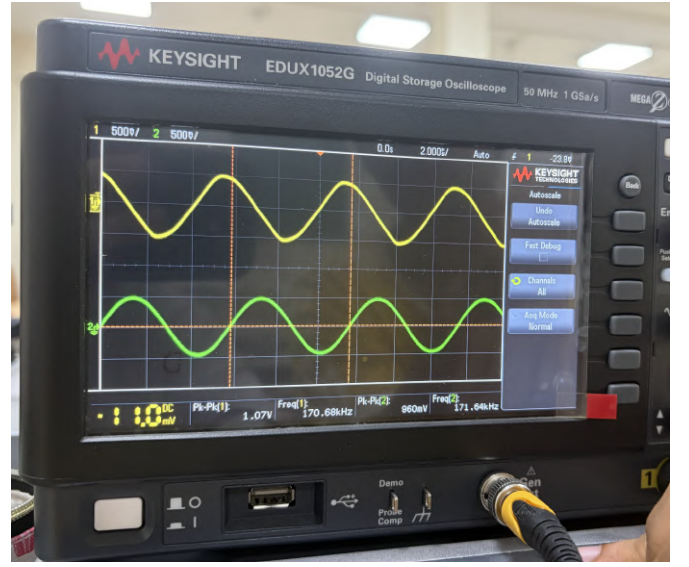
The simulated outputs were sinusoidal and nearly equal in amplitude. The measured values from simulation were approximately 1.05 V_{PP} for v_{OSCI} and 0.99 V_{PP} for v_{OSCO} . The measured oscillation frequency was approximately 169.3 kHz. The time difference between the two waveform peaks was measured as $\Delta t = 1.48 \mu\text{s}$. For a period of approximately $T = 5.91 \mu\text{s}$, the phase difference is

$$\phi = 360^\circ \frac{\Delta t}{T} = 360^\circ \frac{1.48}{5.91} \approx 91^\circ. \quad (15)$$

This is close to the required 90° quadrature relation.

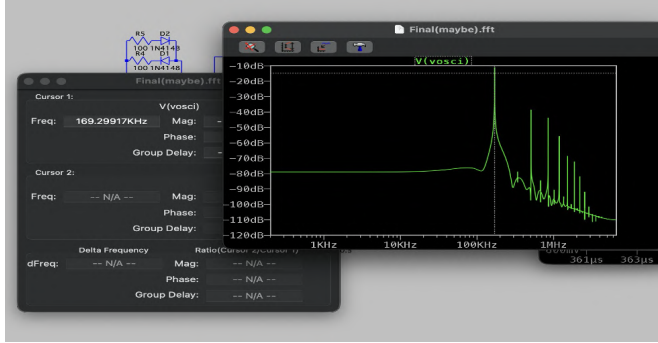


(a) LTspice transient waveforms quadrature oscillator.

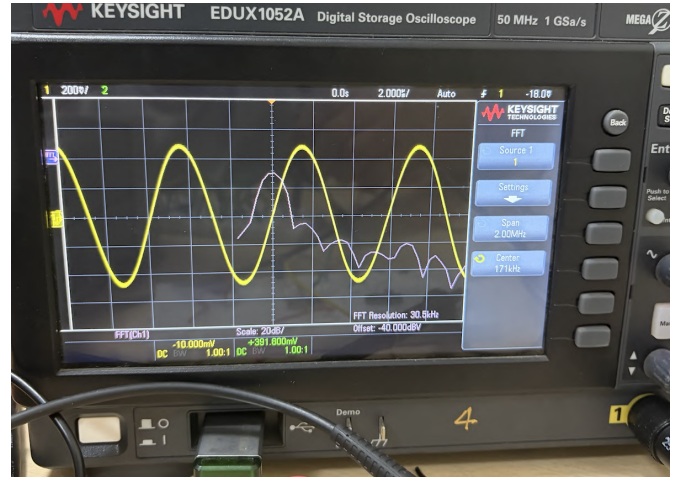


(b) Measured transient waveforms on the hardware circuit. quadrature oscillator.

Fig. 3: Quadrature oscillator transient outputs from simulation and hardware.



(a) LTspice FFT of oscillator output, quadrature oscillator



(b) Measured FFT of oscillator output, quadrature oscillator

Fig. 4: Frequency-domain results of the quadrature oscillator.

D. Laboratory Implementation

The quadrature oscillator was implemented on a breadboard using two UA741 op-amp ICs, passive components, and a $\pm 12\text{V}$ dual supply. In hardware, R was changed from $740\ \Omega$ and $740\ \Omega$ to $806\ \Omega$ because $806\ \Omega$ was the nearest available resistor value.

The measured frequency was approximately 170.68 kHz for the I output and 171.64 kHz for the Q output. The measured peak-to-peak amplitudes were approximately 1.07 V_{PP} and 0.96 V_{PP} respectively. These values are close to the simulation results and meet the intended oscillator requirements.

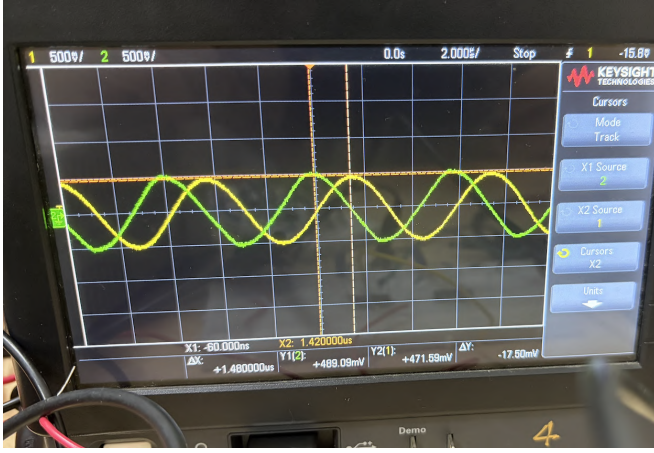


Fig. 5: Oscillator phase-difference measurement with $\Delta t = 1.48 \mu s$.

IV. NMOS SWITCH MIXER DESIGN

A. MOSFET Dimensions and Parasitic Parameters

The NMOS mixer was implemented in LTspice using the provided 180 nm CMOS model. The MOSFET was sized to obtain effective switching and mixing action while maintaining reasonable analog performance. The selected dimensions and parasitic parameters were:

$$L = 0.18 \mu m, \quad W = 1.8 \mu m, \quad (16)$$

$$AD = AS = 8.1 \text{ pm}^2, \quad PD = PS = 8.1 \text{ pm}^2. \quad (17)$$

The LTspice device line was

```
CMOSN L=0.18u W=1.8u AD=8.1p AS=8.1p PD=8.1p PS=8.1p
.include TSMC_180nm.txt
```

B. Mixer Topology

The NMOS transistor was used as a nonlinear switch-based mixer. The oscillator signal was applied to the gate, the input signal was applied through a coupling capacitor, and the output was taken across R_L . The nonlinear device operation generates sum and difference frequency components at the output.

The main component values used in the NMOS mixer are summarized in Table III.

TABLE III: NMOS Mixer Component Values

Component	Value	Purpose
R_{BIAS}	106 k $\blacksquare$	Gate/input biasing
C_C	10 μ F	AC coupling capacitor
R_L	1 k $\blacksquare$	Mixer load
V_{BIAS}	0.9 V	MOS bias voltage
v_{in} amplitude	100 mV	RF input amplitude

C. Mixer Simulation Results

For the mixer simulations, the oscillator frequency was approximately 170 kHz. The input frequency was swept around the oscillator frequency to verify generation of the difference frequency. The tested values were 165 kHz, 168 kHz, 169 kHz, 171 kHz, 172 kHz, and 175 kHz. Therefore, the expected IF

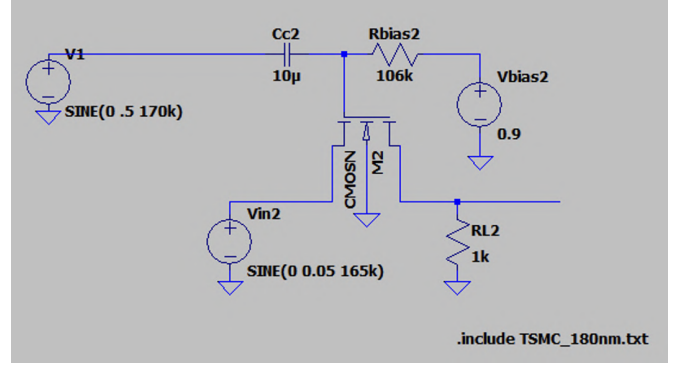


Fig. 6: LTspice schematic of the NMOS switch mixer.

TABLE IV: Expected Mixer Difference Frequencies

f_{OSC}	f_{IN}	Expected $f_{IF} = f_{IN} - f_{OSC} $
170 kHz	165 kHz	5 kHz
170 kHz	168 kHz	2 kHz
170 kHz	169 kHz	1 kHz
170 kHz	171 kHz	1 kHz
170 kHz	172 kHz	2 kHz
170 kHz	175 kHz	5 kHz

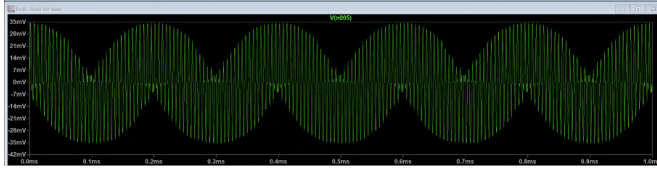
values were 5 kHz, 2 kHz, 1 kHz, 1 kHz, 2 kHz, and 5 kHz respectively.

The FFT results show the expected mixing behavior. Along with high-frequency components near the sum frequency, the output contains a low-frequency component at the difference frequency. This verifies that the NMOS transistor is acting as a nonlinear mixer.

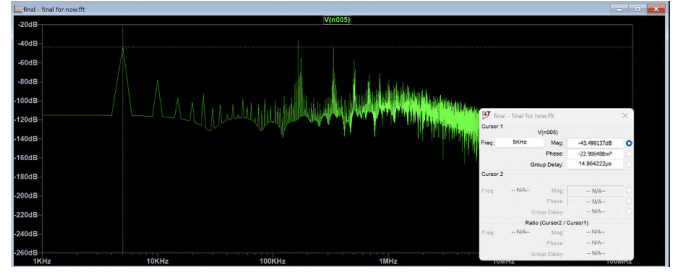
D. Hardware Mixer Results

The mixer was implemented in the laboratory using discrete NMOS transistors and external signal generators. One generator supplied the local oscillator signal and the other supplied the RF input signal. The measured output showed an amplitude-modulated waveform with a visible low-frequency envelope. The observed envelope frequency was approximately 10.98 kHz, corresponding to the difference between the two applied input frequencies in that experiment. The measured peak-to-peak output voltage was approximately 82 mV.

The difference between simulation and lab results is expected because the lab NMOS device has fixed dimensions and parasitic capacitances that differ from the simulated 180 nm CMOS device. Additional mismatch is caused by breadboard parasitics, function generator loading, probe capacitance, and component tolerances.

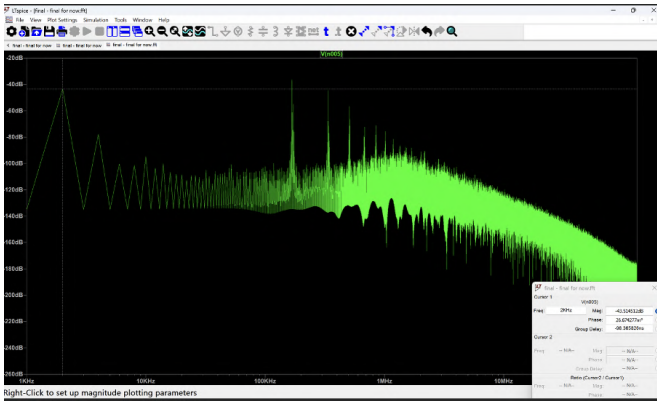


(a) Transient mixer output for $f_{IN} = 165$ kHz.

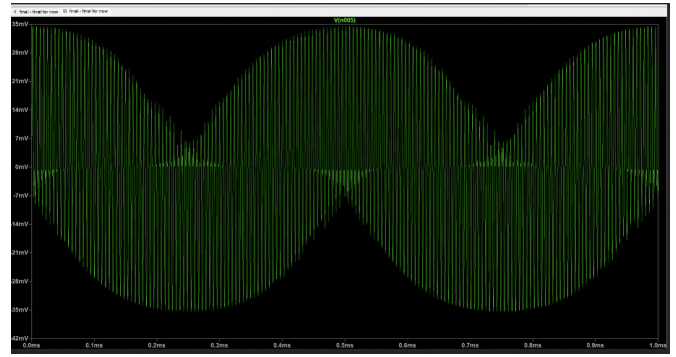


(b) FFT of mixer output for $f_{IN} = 165$ kHz.

Fig. 7: NMOS mixer simulation result for 165 kHz input.

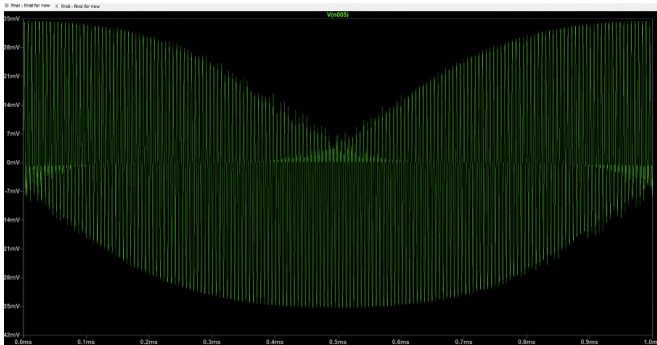


(a) Transient mixer output for $f_{IN} = 168$ kHz.

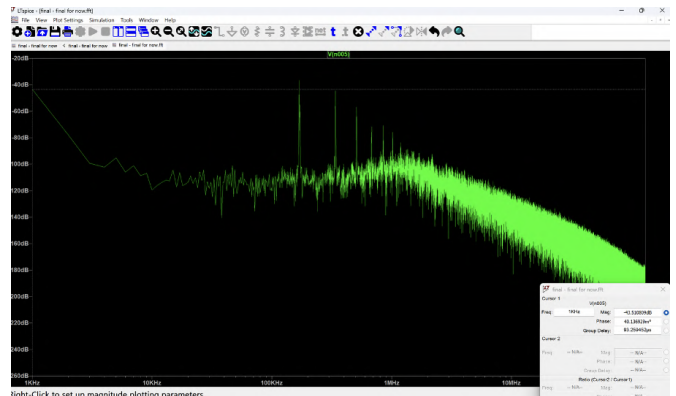


(b) FFT of mixer output for $f_{IN} = 168$ kHz.

Fig. 8: NMOS mixer simulation result for 168 kHz input.

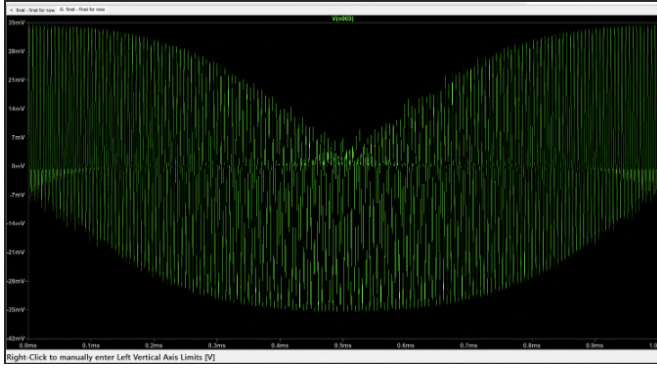


(a) Transient mixer output for $f_{IN} = 169$ kHz.

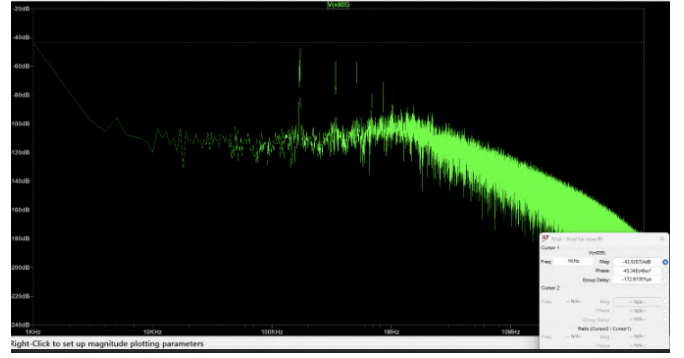


(b) FFT of mixer output for $f_{IN} = 169$ kHz.

Fig. 9: NMOS mixer simulation result for 169 kHz input.

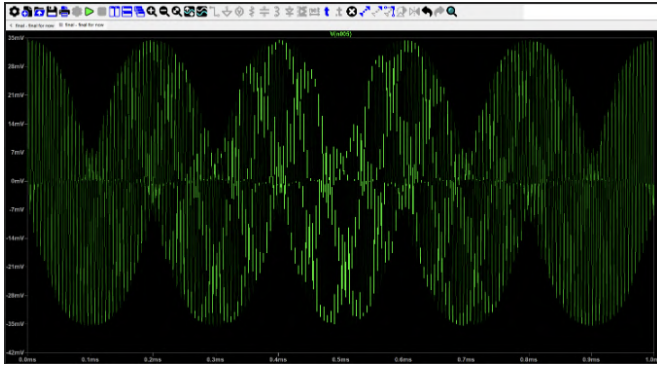


(a) Transient mixer output for $f_{IN} = 171$ kHz.

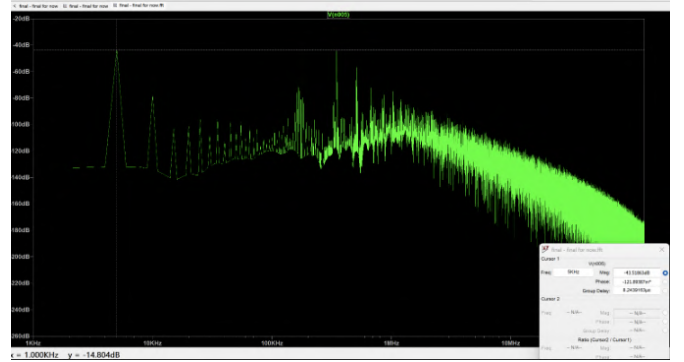


(b) FFT of mixer output for $f_{IN} = 171$ kHz.

Fig. 10: NMOS mixer simulation result for 171 kHz input.



(a) Transient mixer output for $f_{IN} = 175$ kHz.

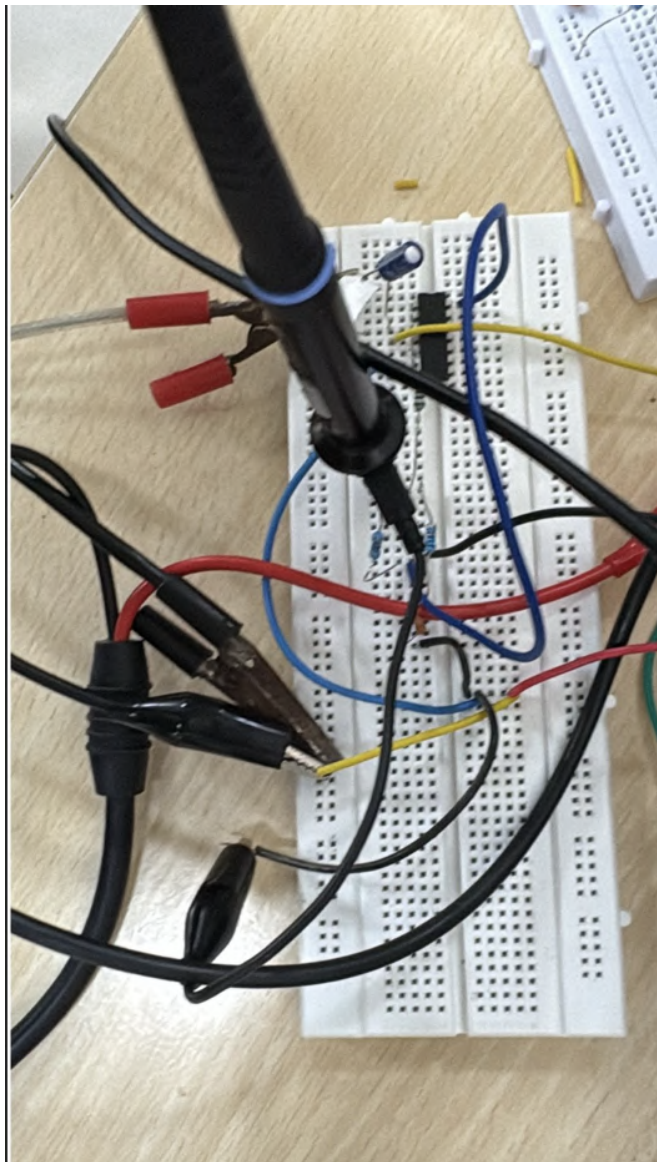


(b) FFT of mixer output for $f_{IN} = 175$ kHz.

Fig. 11: NMOS mixer simulation result for 175 kHz input.



(a) Practical NMOS mixer circuit implemented on breadboard.



(b) Measured output waveform of the practical NMOS mixer.

Fig. 12: Hardware implementation and measured output of the NMOS mixer.

V. LOW-PASS FILTER AND OUTPUT AMPLIFIER DESIGN

Using $R = 5.1 \text{ k}\Omega$ and $C = 10 \text{ nF}$,

$$f_c = \frac{1}{2\pi(5.1 \times 10^3)(10 \times 10^{-9})} \approx 3.12 \text{ kHz}. \quad (19)$$

This is sufficiently close to the required 3 kHz cut-off frequency.

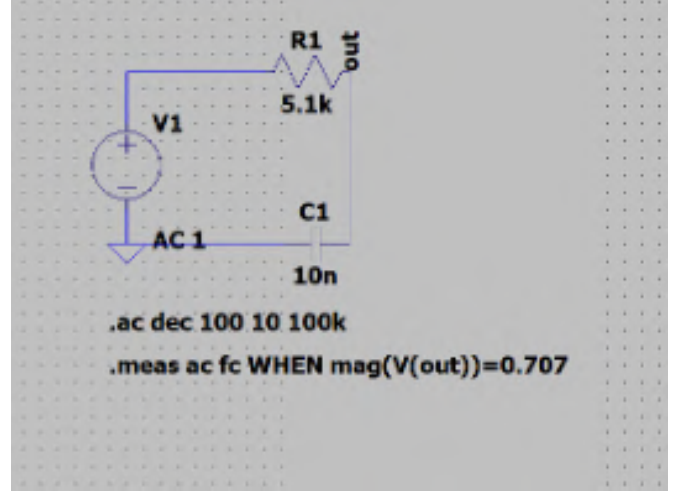


Fig. 13: LTspice schematic of the RC low-pass filter.

A. RC Low-Pass Filter Design

The mixer output contains both the desired low-frequency IF component and undesired high-frequency components around the sum frequency. A first-order RC low-pass filter was designed with a -3 dB cut-off frequency of approximately 3 kHz.

For an RC low-pass filter,

B. AC Analysis

The AC response was simulated in LTspice to verify the low-pass behavior. The magnitude response shows a flat response at low frequencies and attenuation after approximately 3 kHz. The output drops by approximately 3 dB near the designed cut-off frequency.

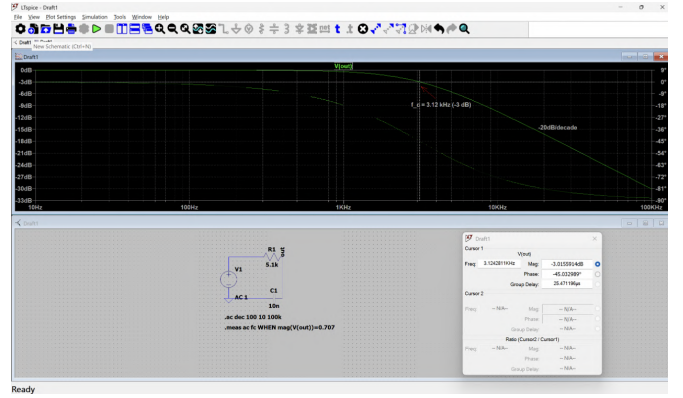
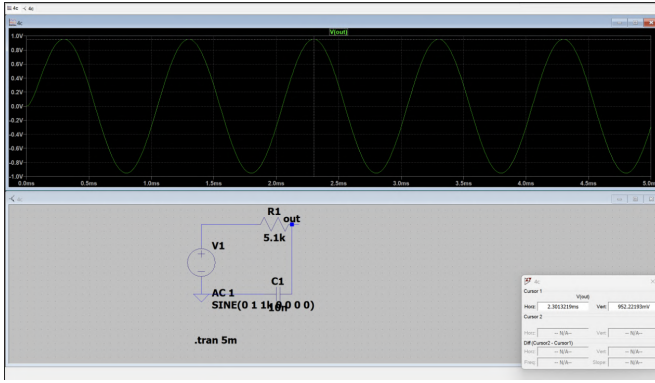


Fig. 14: AC magnitude response of the RC low-pass filter.

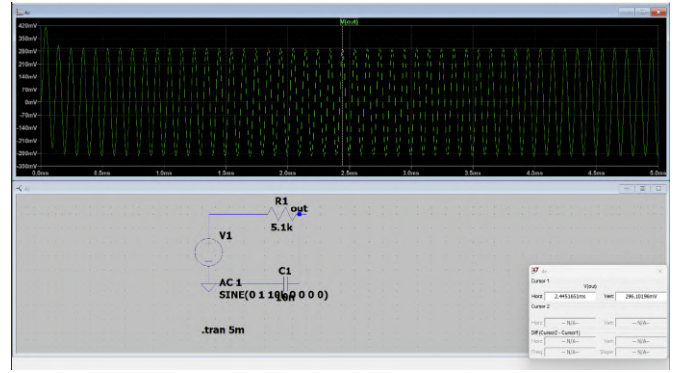
C. Transient Response

The transient response was tested for 1 kHz and 10 kHz sinusoidal inputs. At 1 kHz, the signal is below the cut-off frequency and therefore passes with relatively small attenuation. At 10 kHz, the signal is above the cut-off frequency and is significantly attenuated.

$$f_c = \frac{1}{2\pi RC}. \quad (18)$$

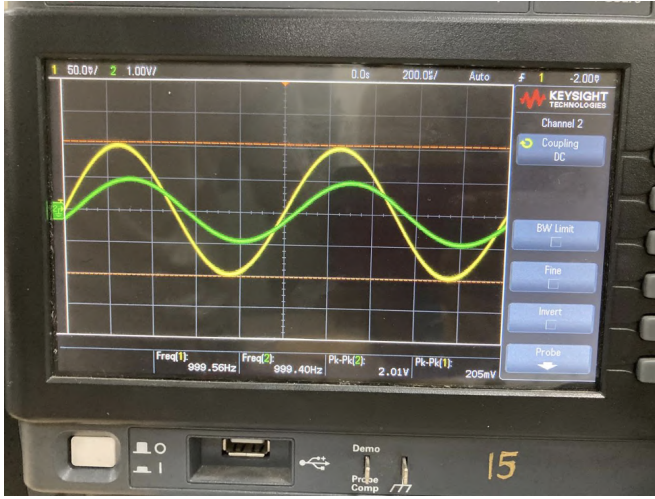


(a) Simulated transient response for 1 kHz input.

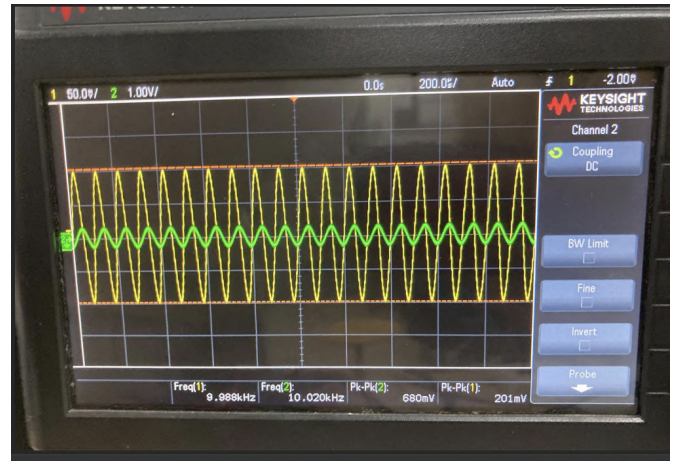


(b) Simulated transient response for 10 kHz input.

Fig. 15: LTspice transient response of the low-pass filter for pass-band and stop-band inputs.



(a) Measured response for 1 kHz input.



(b) Measured response for 10 kHz input.

Fig. 16: Laboratory transient response of the low-pass filter.

D. Mixer with LPF

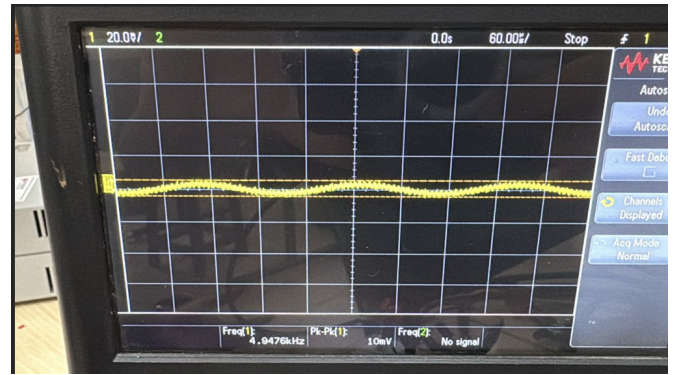


Fig. 17: LTspice setup of the NMOS mixer followed by the RC low-pass filter.

The low-pass filter was connected after the mixer output. This removes the high-frequency sum component and retains the difference-frequency IF output. For input frequencies close to the oscillator frequency, the filtered output becomes a low-frequency sinusoidal signal corresponding to $|f_{IN} - f_{OSC}|$.

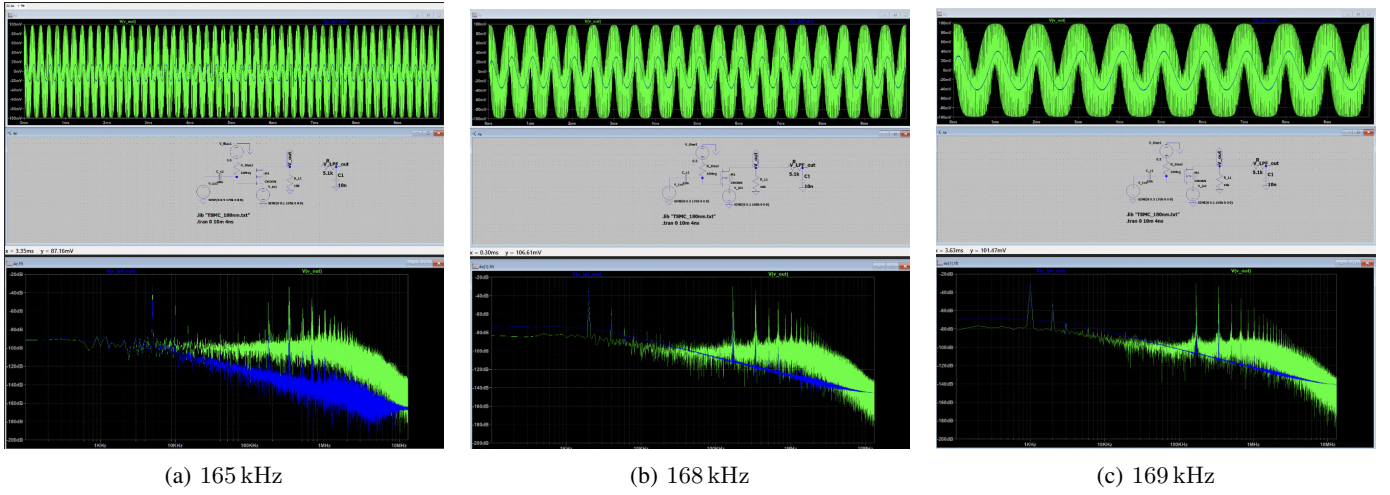


Fig. 18: Mixer plus LPF outputs for input frequencies below and close to the oscillator frequency.

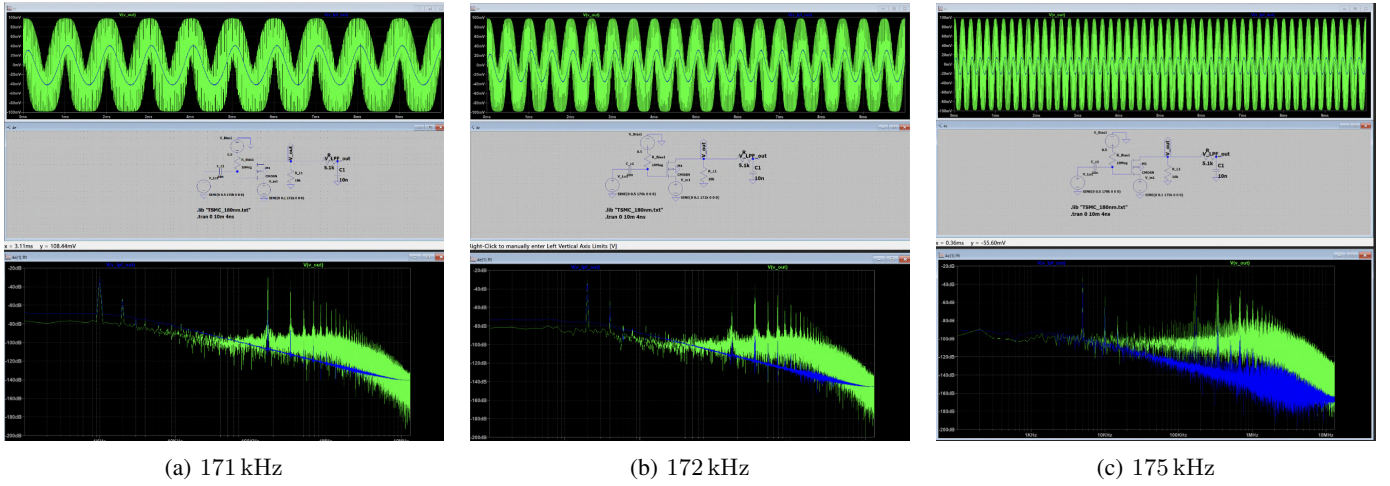
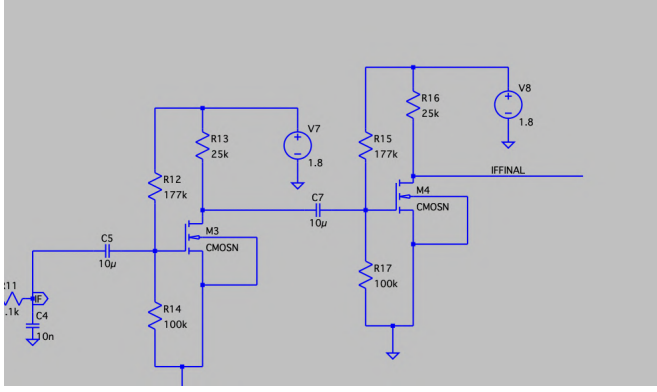


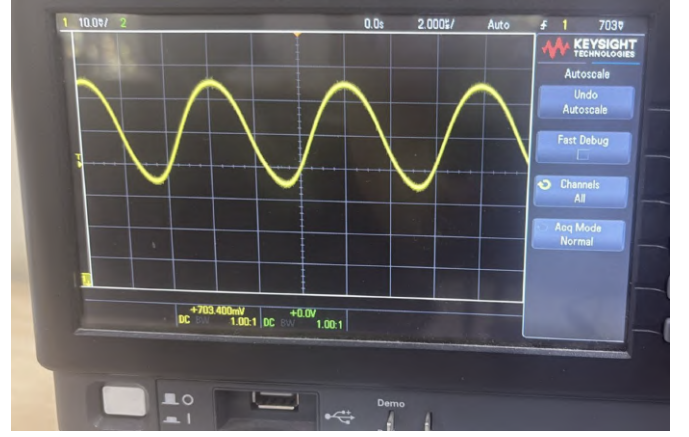
Fig. 19: Mixer plus LPF outputs for input frequencies above the oscillator frequency.

E. Final Amplifier Stage

A MOS amplifier stage was added after the low-pass filter to obtain the final required output amplitude. The target final output amplitude was approximately 200 mV peak, corresponding to 400 mV_{PP}, or 250 mV peak, corresponding to 500 mV_{PP}, depending on the allocated specification. In the complete circuit, the final output stage used biasing resistors and a supply of approximately 1.8 V. The amplifier gain was tuned so that the final IF output amplitude was within the required range without using an op-amp for signal amplification.



(a) MOS amplifier schematic used after the low-pass filter.



(b) Amplifier output showing approximately $7\times$ gain.

Fig. 20: Final amplifier stage used for IF output conditioning.

VI. COMPLETE QUADRATURE DOWN CONVERTER PROTOTYPE

A. Complete Circuit Description

The complete circuit combines the quadrature oscillator, two NMOS mixer paths, two low-pass filters, and two final amplifier paths. The input signal v_{in} is fed to both mixer paths. The I mixer is driven by v_{OSCI} , while the Q mixer is driven by v_{OSCQ} . Since the local oscillator signals differ by approximately 90° , the final filtered outputs also preserve the I-Q phase relation.

B. Component Summary for Complete Circuit

C. Simulation of Complete Circuit

Transient simulations were carried out for the complete QDC. The plotted nodes include the input signal, oscillator I and Q signals, mixer IF outputs, and final filtered IF outputs. The oscillator waveforms show the expected quadrature relation, and the final I and Q outputs show the down-converted low-frequency IF component.

D. Complete Hardware Prototype Results

The complete circuit was realized by connecting all blocks on the breadboard. The measured signals included v_{in} , v_{OSCI} , v_{OSCQ} , v_{IFI} , v_{IFQ} , $v_{IFFINAL,I}$, and $v_{IFFINAL,Q}$. The oscillator outputs retained the quadrature relationship, while the final outputs showed the expected low-frequency IF waveform after filtering and amplification. The I and Q paths were measured separately using a single mixer-LPF-amplifier chain with switched output, as advised by the lab instructor.

E. FFT-Based Phase Extraction of Final I and Q Outputs

The LTSpice FFT cursor was used to extract the phase of the final I and Q outputs directly from their complex spectra. Cursor 1 was placed at the IF magnitude peak of $V(iffinal)$ and separately at the peak of $V(qffinal)$. The readings at matched IF frequencies are summarised in Table VI.

TABLE VI: FFT Cursor Phase Readings for Final I and Q Outputs

Signal	Frequency	Magnitude	Phase
$V(iffinal)$	4.883 kHz	-18.403 dB	-151.00°
$V(qffinal)$	4.811 kHz	-18.467 dB	-240.96°

The phase difference between the two outputs is

$$\Delta\phi = |\phi_I - \phi_Q| = |-151.00^\circ - (-240.96^\circ)| \approx 89.96^\circ \approx 90^\circ. \quad (20)$$

The nearly equal magnitudes confirm balanced I and Q path gains, with less than 0.07 dB mismatch. The residual error of 0.04° is within simulation numerical precision, confirming that the complete QDC produces near-perfect quadrature outputs.

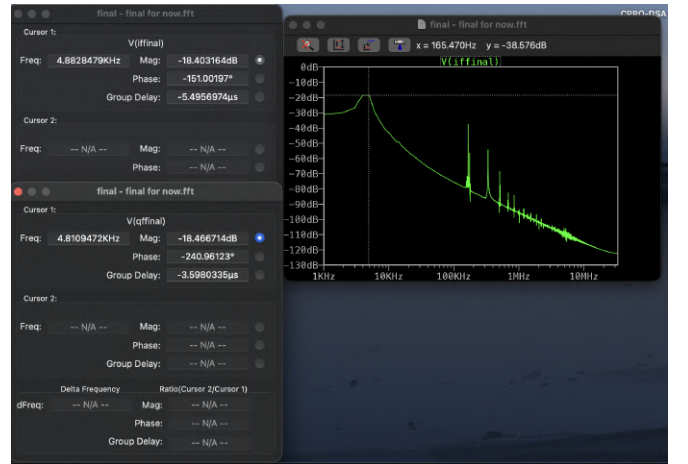


Fig. 29: LTSpice FFT cursor readouts for $V(iffinal)$ (Phase = -151.00°) and $V(qffinal)$ (Phase = -240.96°), giving $\Delta\phi \approx 90^\circ$.

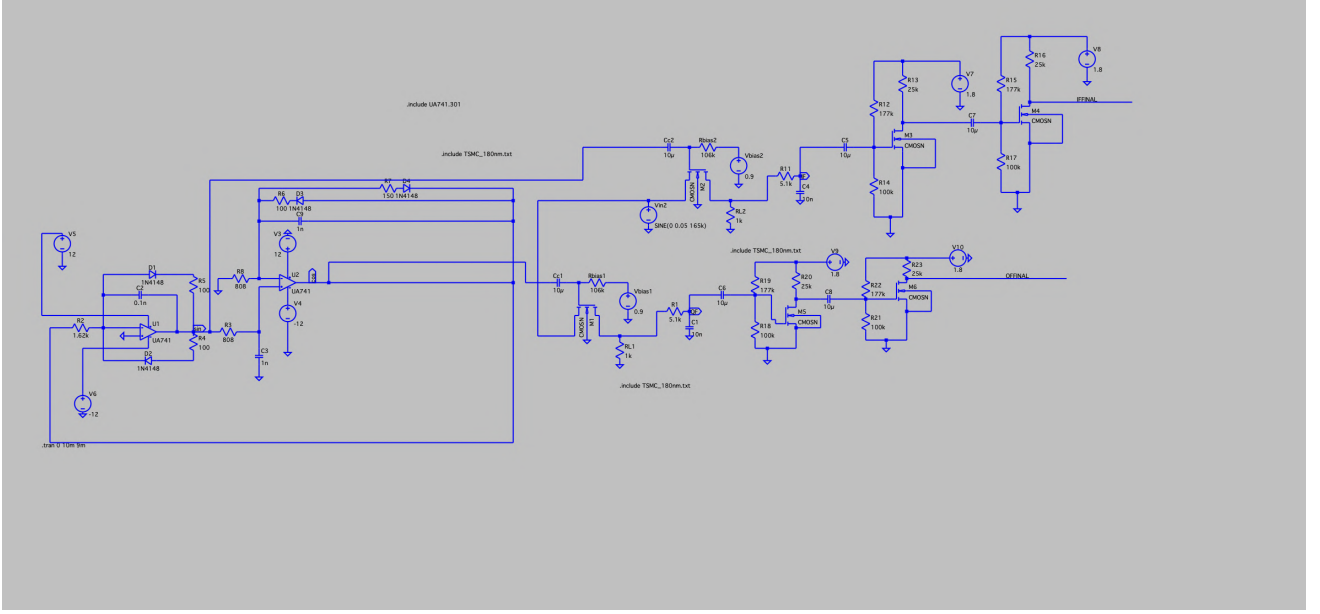


Fig. 21: Complete LTspice schematic of the quadrature down converter prototype.

TABLE V: Component and Supply Values Used in Complete QDC Prototype

Block	Component	Value	Purpose
Oscillator	UA741 supply	± 12 V	Op-amp supply
Oscillator	$R_1, R_2, R_3, C_1, C_2, C_3$	As in Table I	Quadrature generation
Mixer I/Q	R_{BIAS1}, R_{BIAS2}	106 k $\square$	NMOS biasing
Mixer I/Q	C_{C1}, C_{C2}	10 μ F	AC coupling
Mixer I/Q	R_{L1}, R_{L2}	1 k $\square$	Mixer load
Mixer I/Q	V_{BIAS1}, V_{BIAS2}	0.9 V	Gate/source bias
LPF I/Q	R	5.1 k $\square$	LPF resistor
LPF I/Q	C	10 nF	LPF capacitor
Amplifier I/Q	Supply	1.8 V	MOS amplifier supply
Amplifier I/Q	Bias resistors	177 k $\square$, 100 k $\square$, 25 k $\square$	MOS biasing network
Input	v_{in}	100 mV	RF input signal

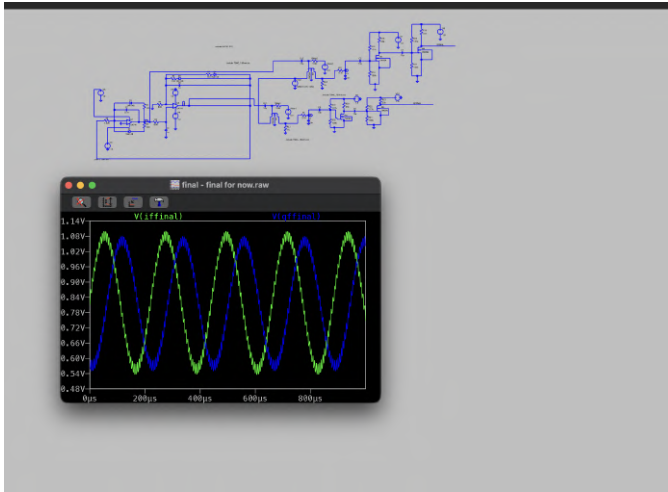


Fig. 22: Transient output of the complete QDC.

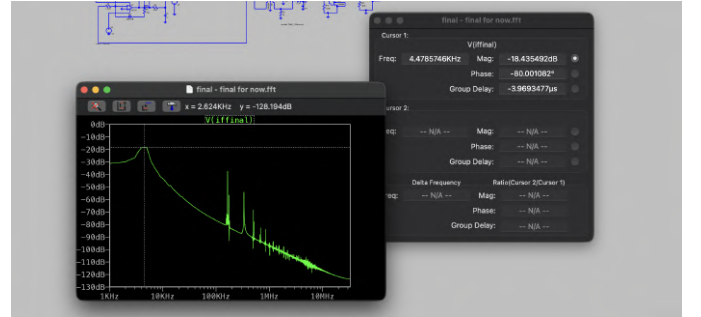


Fig. 23: FFT output of the complete QDC.

F. Simulation and Measurement Comparison

TABLE VII: Performance Summary and Comparison

Parameter	Simulated	Measured	Remarks
Oscillator frequency	169.3 kHz	170.68 kHz / 171.64 kHz	Close
Oscillator amplitude, I	1.05 V _{pp}	1.07 V _{pp}	In range
Oscillator amplitude, Q	0.99 V _{pp}	0.96 V _{pp}	Slightly low
I-Q phase difference	91 $^{\circ}$	$\approx 91^{\circ}$	Quadrature
FFT phase difference	89.96 $^{\circ}$	—	Confirmed
Input frequency	165 kHz	165 kHz	Generator
Mixer output amplitude	11 mV	10 mV _{pp}	Gain input

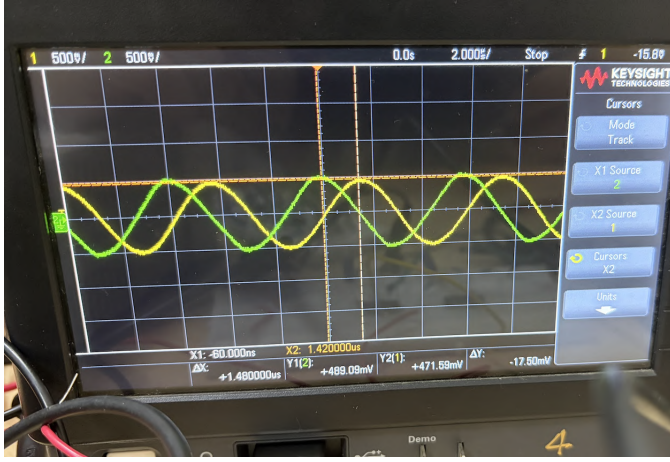


Fig. 24: Transient v_{OSCQ} and v_{OSCI} .

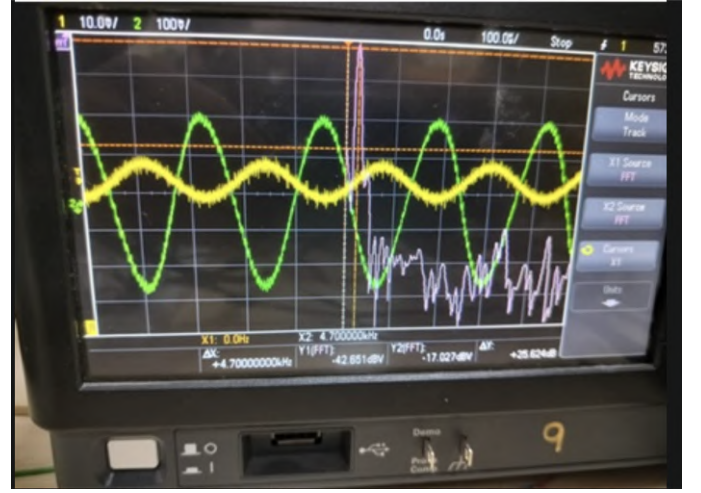


Fig. 27: Final IF output $v_{IFFINAL}$.

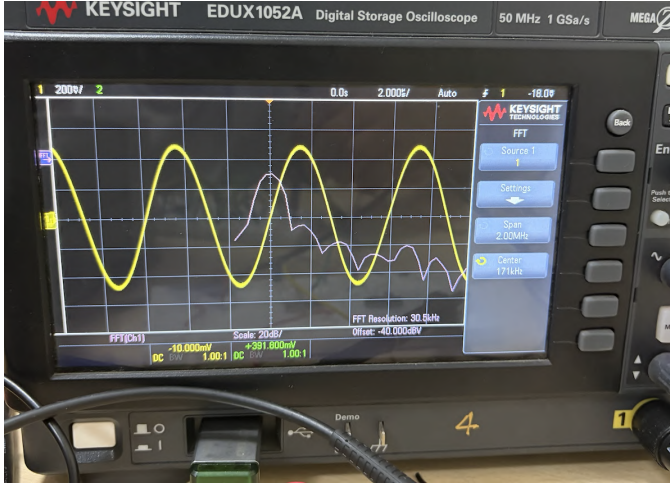


Fig. 25: FFT of v_{OSCQ} and v_{OSCI} .

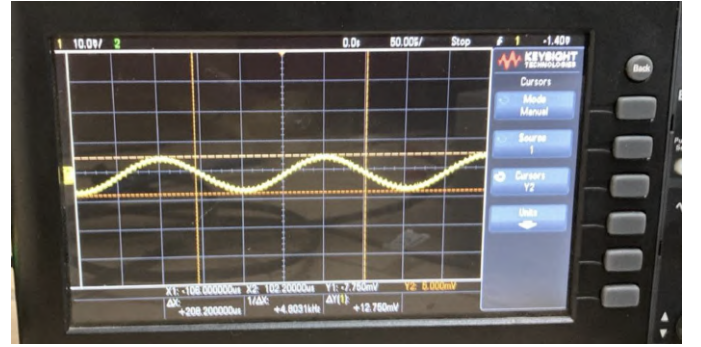


Fig. 28: IF output v_{IF} .

VII. DISCUSSION

The QDC prototype demonstrates quadrature down conversion using separate oscillator, mixer, filter, and gain stages. The oscillator generates nearly quadrature local oscillator signals, the NMOS mixer produces sum and difference components, and the low-pass filter retains the desired difference-frequency IF component. The final amplifier increases the filtered IF amplitude without using an op-amp as a signal amplifier. FFT-based phase extraction confirmed an I-Q phase separation of 89.96° , validating the quadrature operation of the complete system.

VIII. CONCLUSION

A complete quadrature down converter was designed and verified through LTspice simulations and laboratory measurements. The quadrature oscillator produced approximately 1 V_{pp} outputs near 170 kHz with a phase difference close to 90° . The NMOS mixer generated the expected difference-frequency IF components for input frequencies around the oscillator frequency. FFT phase extraction of the final outputs yielded $\Delta\phi = 89.96^\circ \approx 90^\circ$, confirming near-perfect quadrature. The complete QDC prototype produced in-phase and quadrature IF outputs, validating the design flow from individual blocks to full-system implementation.

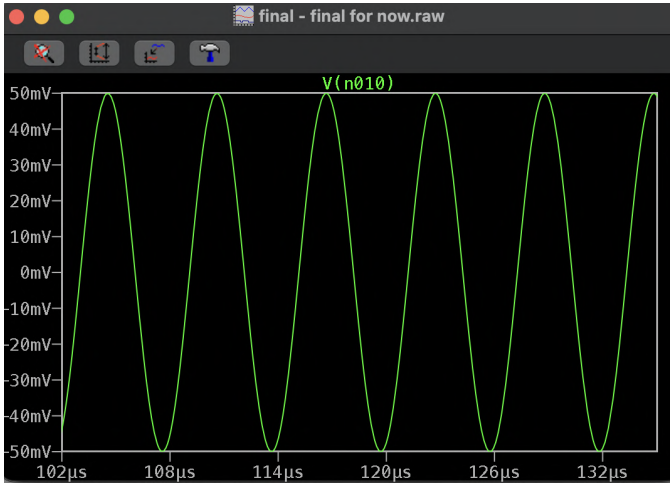


Fig. 26: Input signal v_{in} .

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